

Chapter 3

PROPOSED SYSTEM

The Proposed mobile application operates only on the basis of location; there is no requirement to rely on a specific dealer. We can forecast when the level of the water will fall using machine learning. The customer can then contact the admin, who will make the required arrangements to supply water. Anyone in need of water can make a request for it from the local vendors based on their needs. The user's request alerts the supplier, who then meets the user's water need appropriately. The simplicity, time efficiency, user friendliness, and cost effectiveness of the suggested mobile application are its benefits.

3.1 Data Preprocessing

Data preparation is an essential stage in creating an Android app for water supply. It entails setting up the gathered data for analysis and display by organising and preparing it. Data is first collected from a variety of sources, including sensors, user inputs, and other databases that are pertinent to the application. To deal with errors, inconsistencies, and missing numbers, data cleansing is done next. To guarantee data quality, methods like data imputation, duplicate elimination, and formatting correction are used. Data integration is the process of combining information from several sources to produce a comprehensive dataset that has all necessary information for analysis. Data transformation is putting the gathered information into a format that will be useful for additional research or visualisation. During this step, categorical variables may be subjected to scaling, normalisation, and encoding. By identifying the dataset's most pertinent characteristics, feature selection or extraction lowers the dataset's dimensionality and boosts the performance of the model. Data aggregation compiles or combines data at various granularities to produce insightful summaries.